B3.2 Transport in plants (HL)

NAME	DATE	TOTAL
		_____ / 20

About 20 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

5 MARKS

1 **DEFINE** translocation. [B3.2.18] [1]

2 **STATE** the name for drops of xylem sap pushed out at the edges of leaves by root pressure. [B3.2.17] [1]

3 **STATE** two features of companion cells. [B3.2.18] [2]

4 **NAME** one part of a plant that acts as a sink. [B3.2.18] [1]

Section B: Understanding

9 MARKS

5 **EXPLAIN** how root pressure is generated. [B3.2.17] [3]

6 **EXPLAIN** how sieve tube elements are adapted for the flow of sap. [B3.2.18] [3]

7 **SUGGEST** why root pressure matters more to a deciduous tree in early spring than in summer. [B3.2.17] [3]

Section C: Working with evidence

6 MARKS

All data in this section are invented for practice. The patterns are realistic.

8 [B3.2.17]

The shoots of two groups of tomato plants were cut off just above the soil, and the sap that came out of each stump was collected. After 1 hour, a respiratory inhibitor was added to the water around the roots of one group. The table shows the mean total volume of sap collected.

TIME / H	0	1	2	3	4
NO INHIBITOR / CM^3	0	0.42	0.85	1.26	1.70
INHIBITOR ADDED AT 1 H / CM^3	0	0.40	0.55	0.58	0.59

a CALCULATE the mean rate of sap flow from the plants with no inhibitor over the 4 hours. [1]

b DESCRIBE the effect of the inhibitor. [2]

c EXPLAIN the effect of the inhibitor. [3]